R PROGRAMMING

Name : SHANE MAC S

Reg No : 192211511

EXPERIMENT-1

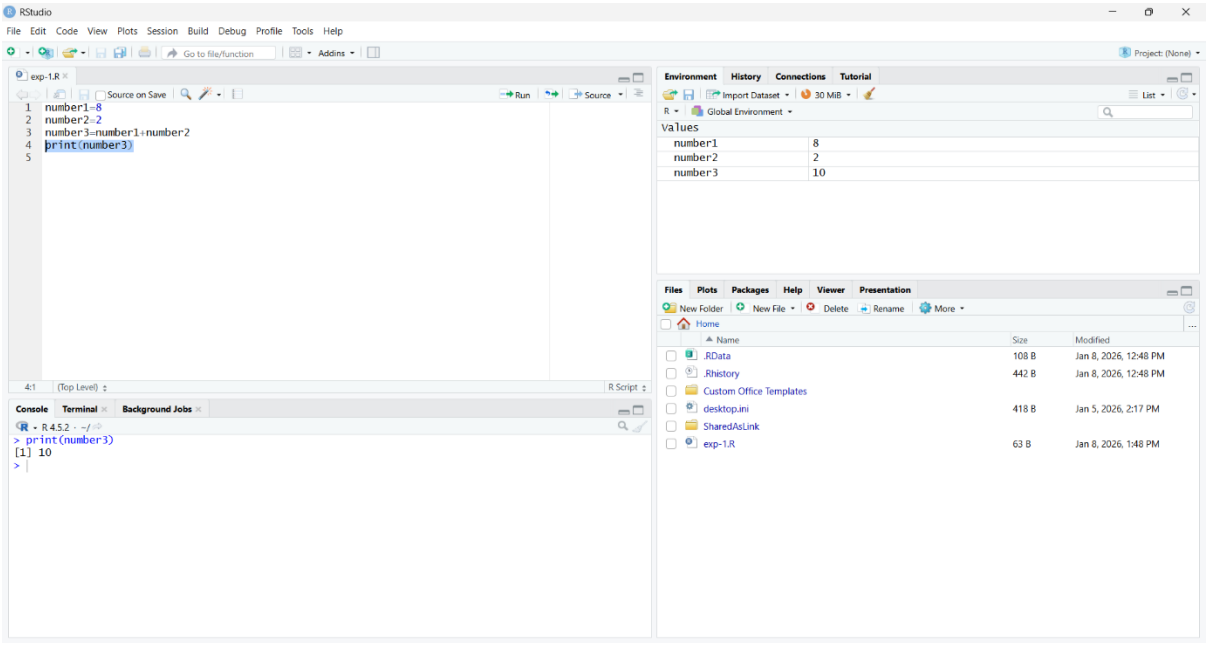
AIM:

To prove the program for addition using R-tool.

PROGRAM:

```
num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))
sum_result <- num1 + num2
print(sum_result)
```

OUTPUT:



RESULT:

The basic program addition is executed successfully.

EXPERIMENT-2

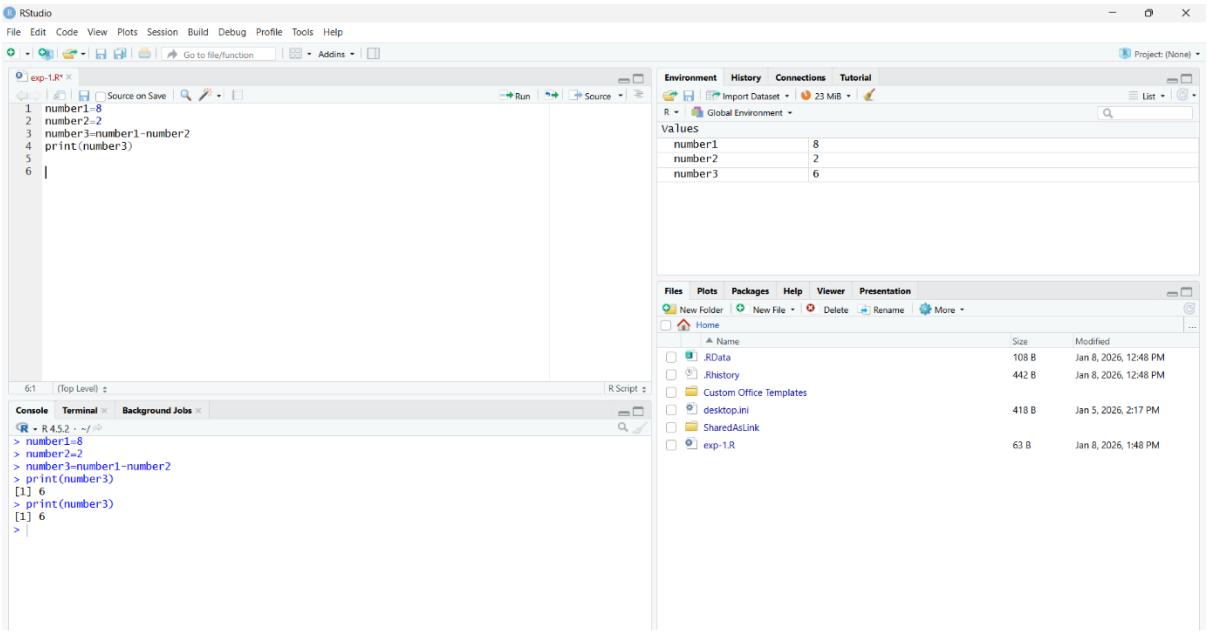
AIM:

To prove the program for subtraction using R-tool.

PROGRAM:

```
num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))
difference <- num1 - num2
print(difference)
```

OUTPUT:



RESULT:

The basic program subtraction is executed successfully.

EXPERIMENT-3

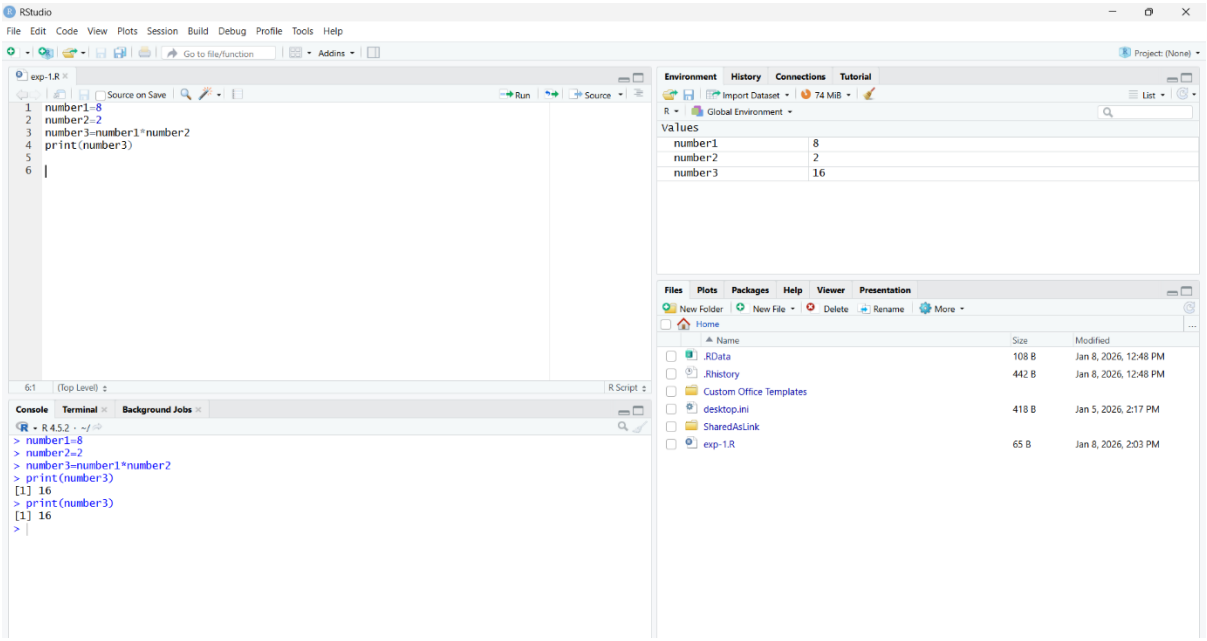
AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

```
num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))
product <- num1 * num2
print(product)
```

OUTPUT:



RESULT:

The basic program multiplication is executed successfully.

EXPERIMENT-4

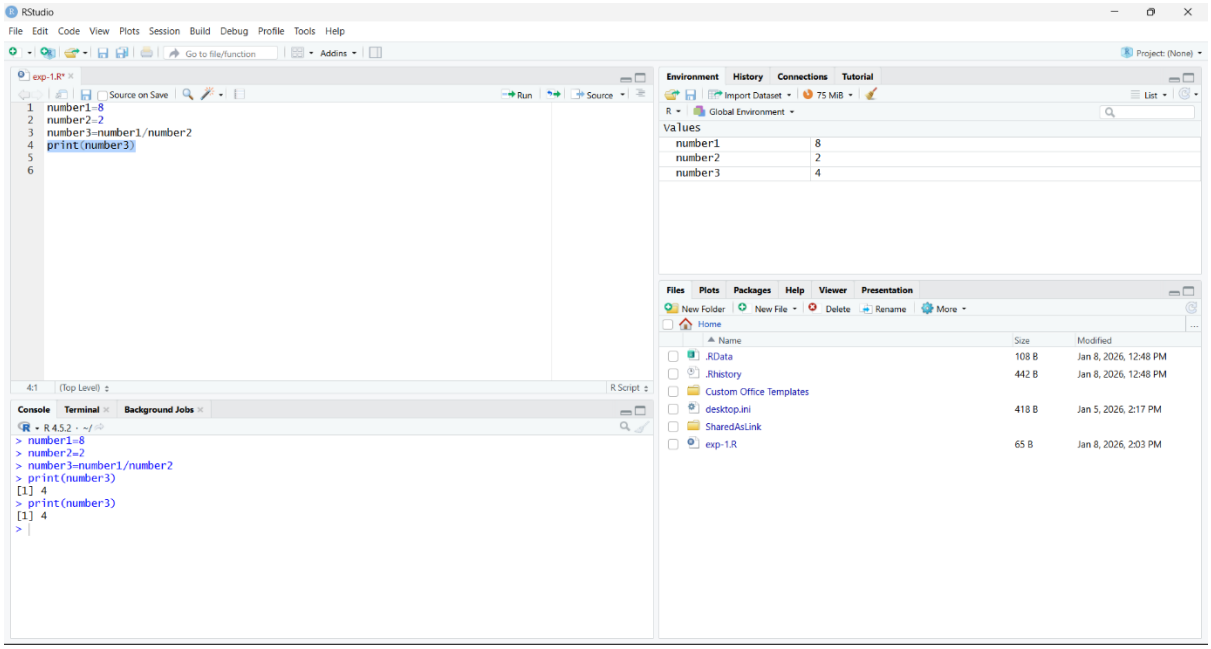
AIM:

To prove the program for division using R-tool.

PROGRAM:

```
num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))
if (num2 == 0) {
  stop("Division by zero is not allowed.")
}
quotient <- num1 / num2
print(quotient)
```

OUTPUT:



RESULT:

The basic program division was executed successfully.

EXPERIMENT-5

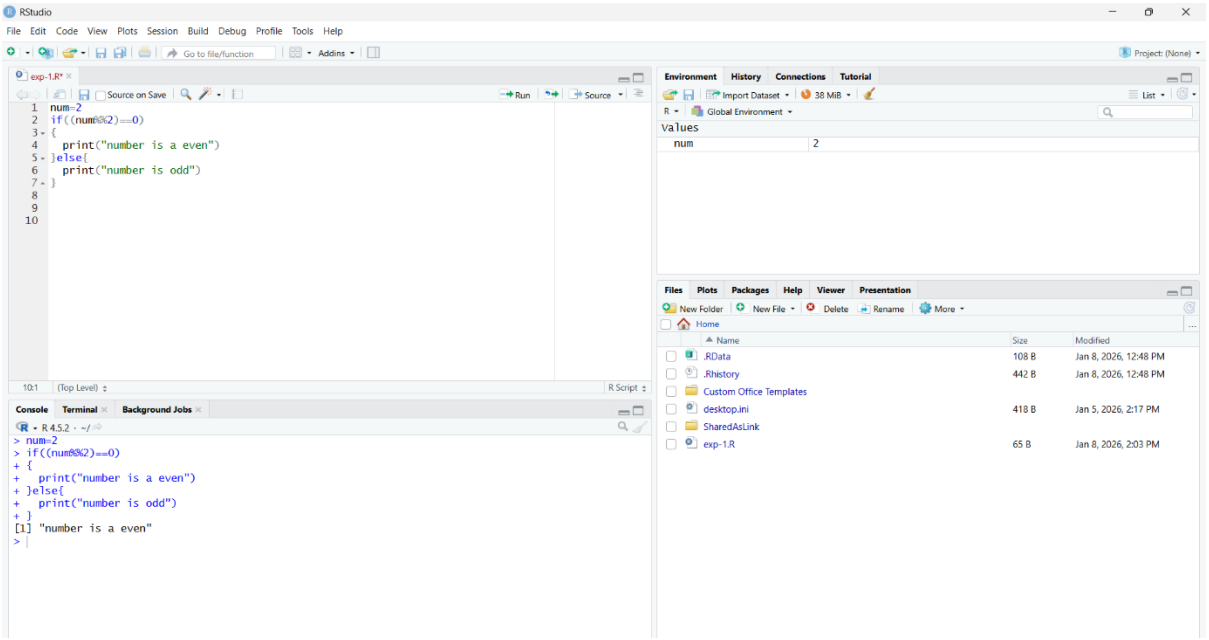
AIM:

To write the program for odd or even using R-tool.

PROGRAM:

```
num <- as.integer(readline(prompt = "Enter an integer: "))
if (num %% 2 == 0) {
  print("Number is even")
} else {
  print("Number is odd")
}
```

OUTPUT:



RESULT:

The basic program odd or even was executed successfully.

EXPERIMENT-6

AIM:

To write the program for mean,median,mode.

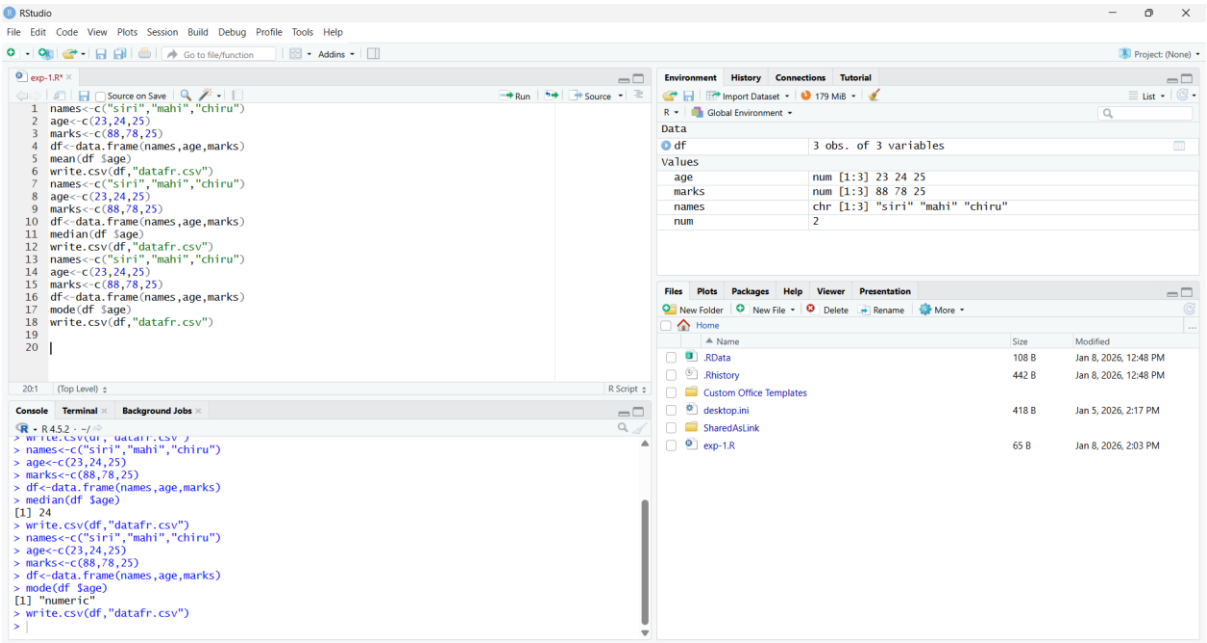
PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)

# Mean and median of age
mean(df$age)
median(df$age)

# Statistical mode (returns NA when every value occurs once)
get_mode <- function(x) {
  tab <- table(x)
  if (max(tab) == 1) return(NA)
  as.numeric(names(tab)[tab == max(tab)][1])
}
get_mode(df$age)
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

The central tendency and measure of dispersion is executed successfully.

EXPERIMENT-7

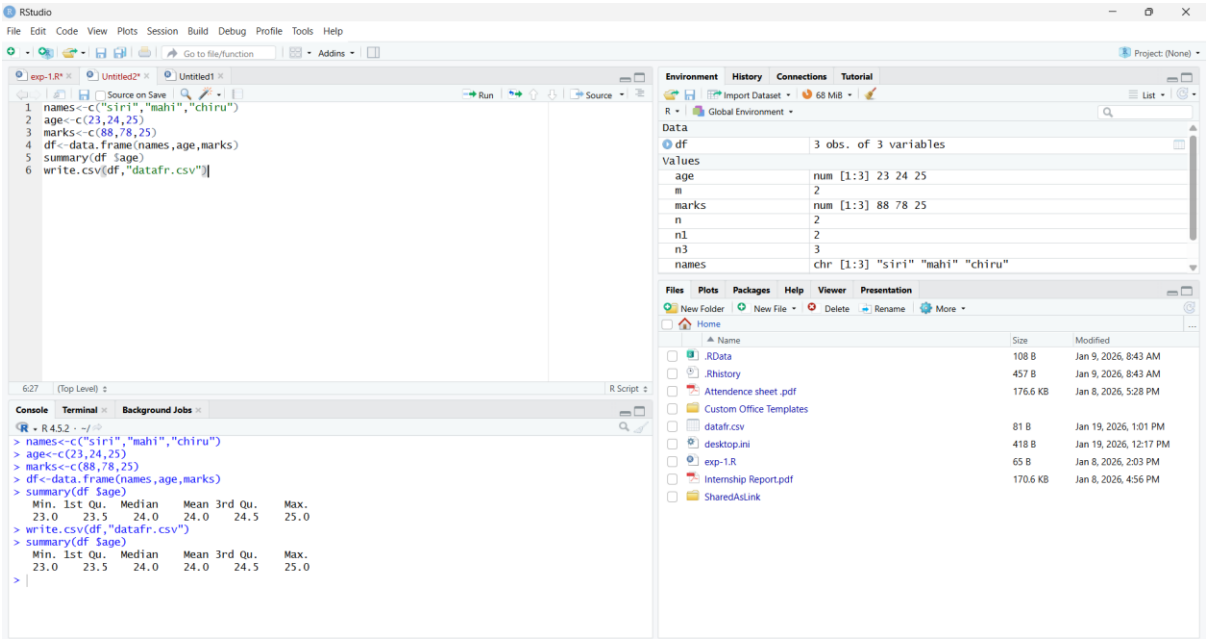
AIM:

To write the program for summary using R-tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
summary(df$age)
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

The central tendency and measure of dispersion is executed successfully.

EXPERIMENT-8

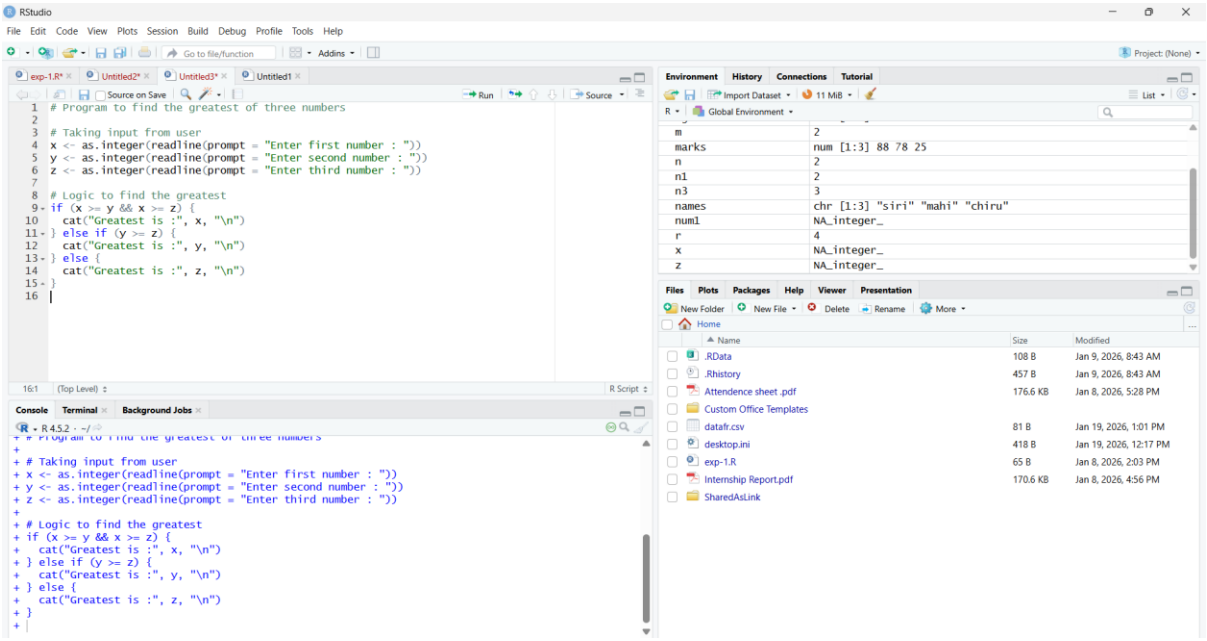
AIM:

To write the program for the greatest among three numbers.

PROGRAM:

```
x <- as.numeric(readline(prompt = "Enter the first number: "))
y <- as.numeric(readline(prompt = "Enter the second number: "))
z <- as.numeric(readline(prompt = "Enter the third number: "))
numbers <- c(x, y, z)
cat("Greatest number is:", max(numbers), "\n")
```

OUTPUT:



RESULT:

The greatest among the three numbers was executed successfully.

EXPERIMENT – 9

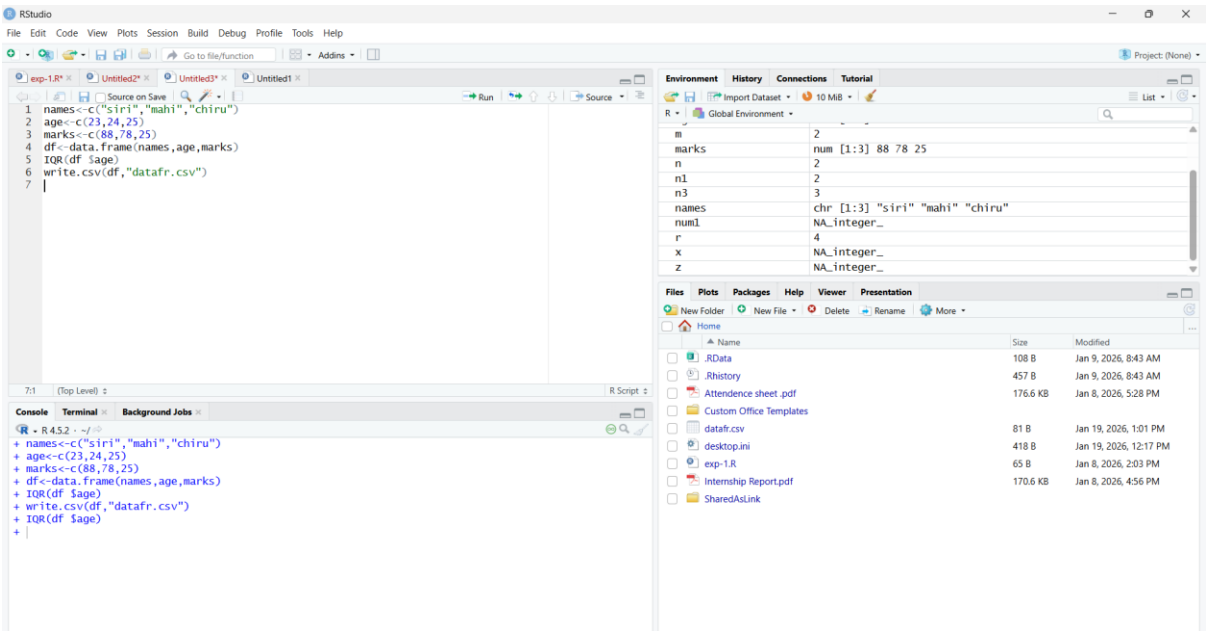
AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
IQR(df$age)
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

The program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT-10

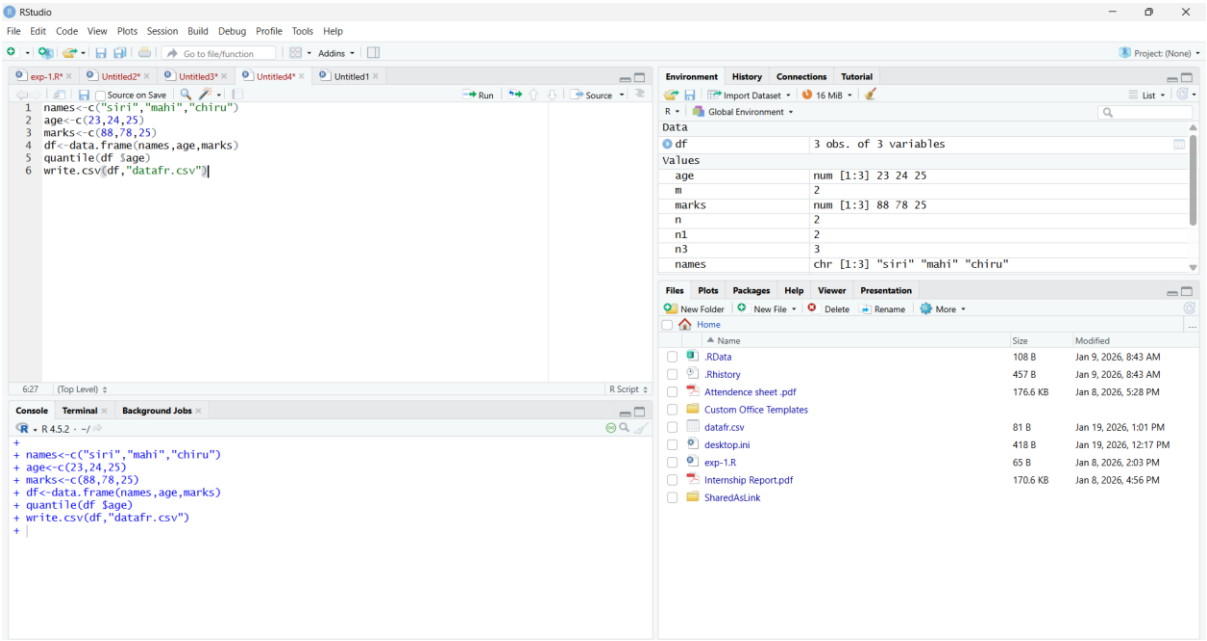
AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
quantile(df$age)
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

The program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-11

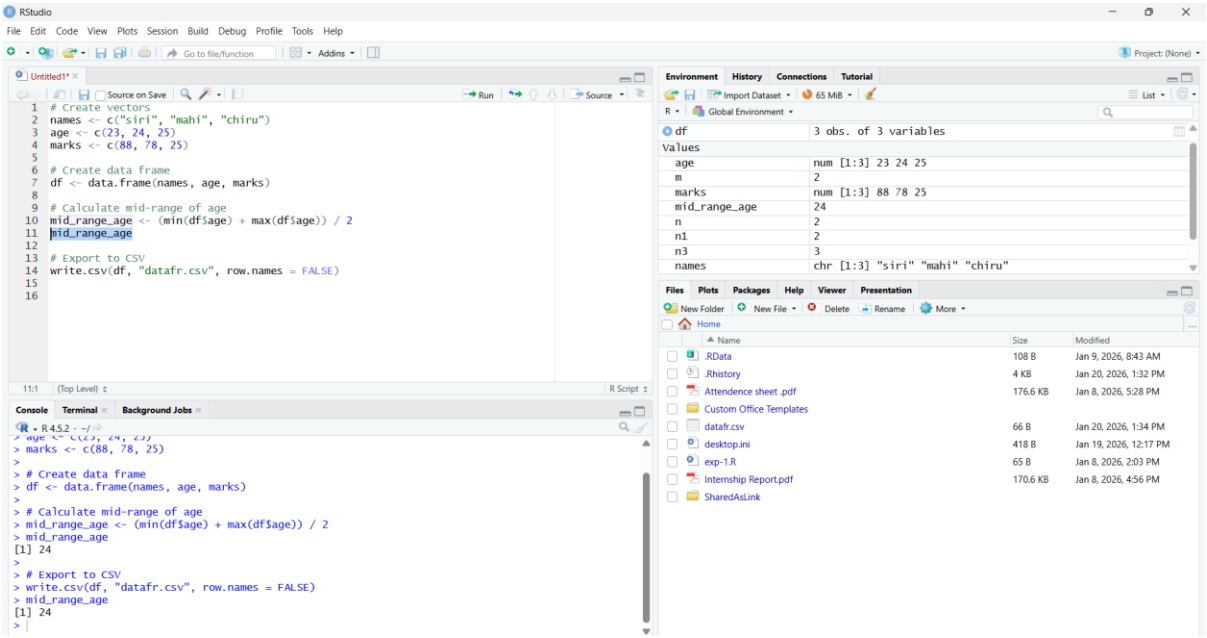
AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
mid_range <- (min(df$age) + max(df$age)) / 2
print(mid_range)
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

The program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-12

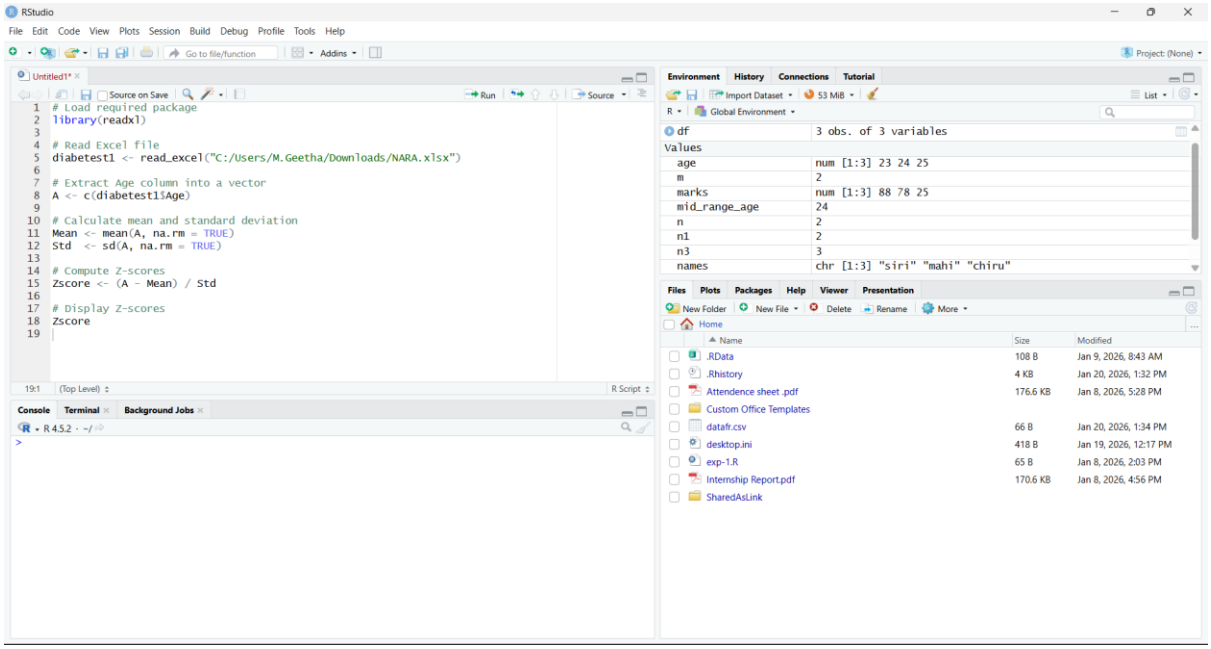
AIM:

To write the program for Z-score normalization using R-tool.

PROGRAM:

```
# Install once if needed: install.packages("readxl")
library(readxl)
diabetest1 <- read_excel("NARA.xlsx")  # Keep the file in the working directory
A <- as.numeric(diabetest1$Age)
Mean <- mean(A, na.rm = TRUE)
Std <- sd(A, na.rm = TRUE)
Zscore <- (A - Mean) / Std
print(Zscore)
```

OUTPUT:



RESULT:

The Z-score normalization using R tool was executed successfully.

EXPERIMENT-13

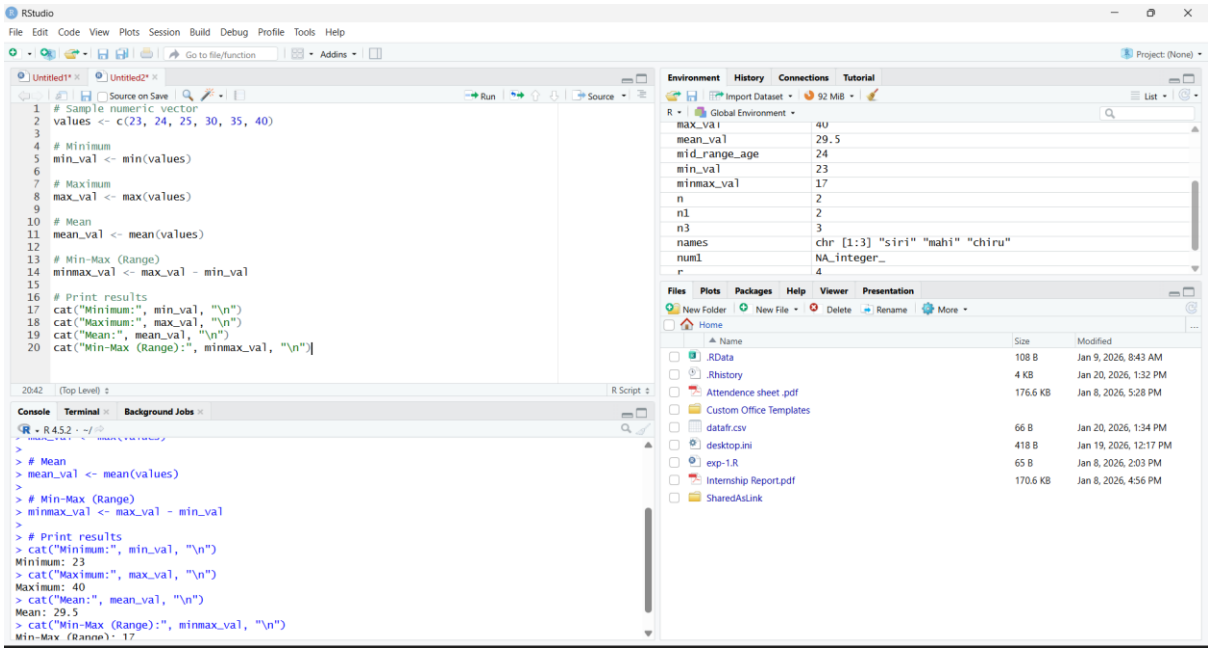
AIM:

To write the program for the minimum,maximum,mean and minmax using r-TOOL

PROGRAM:

```
# Install once if needed: install.packages("readxl")
library(readxl)
diabetest1 <- read_excel("NARA.xlsx")  # Keep the file in the working directory
A <- as.numeric(diabetest1$Age)
Mean <- mean(A, na.rm = TRUE)
Minimum <- min(A, na.rm = TRUE)
Maximum <- max(A, na.rm = TRUE)
MinMax <- (A - Minimum) / (Maximum - Minimum)
cat("Mean:", Mean, "\n")
cat("Minimum:", Minimum, "\n")
cat("Maximum:", Maximum, "\n")
print(MinMax)
```

OUTPUT:



RESULT:

The program for min,max,minmax,mean was executed successfully.

EXPERIMENT-14

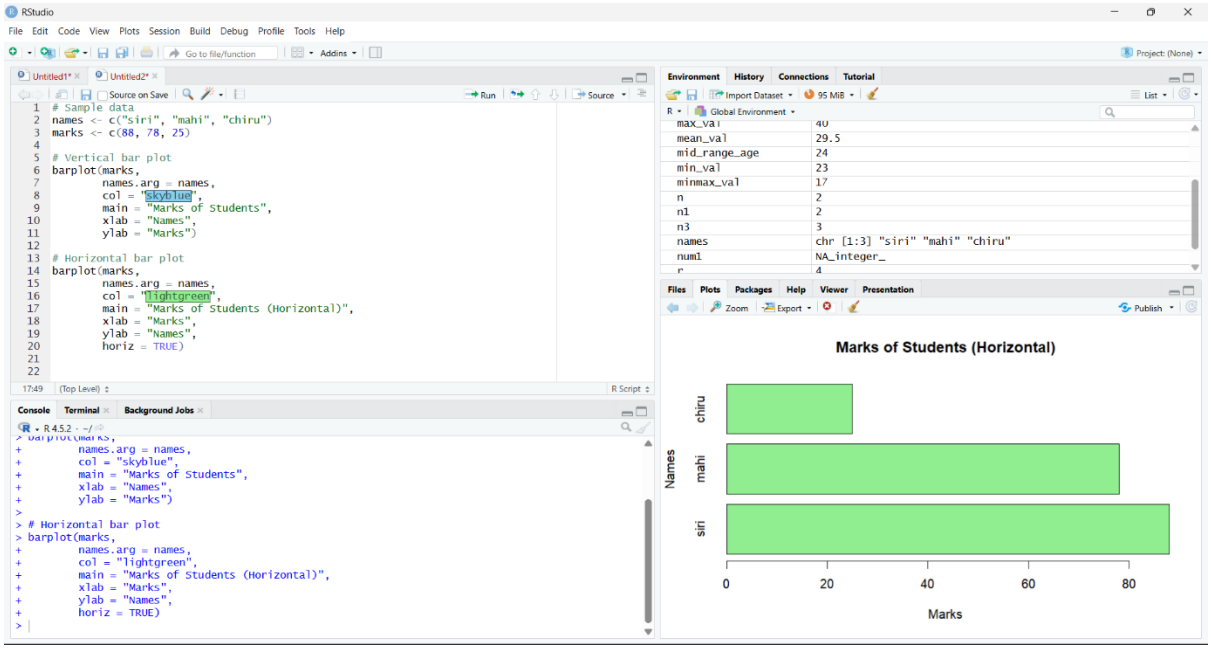
AIM:

To draw the bar plot and horizontal bar using R-tool.

PROGRAM:

```
marks <- c(55, 67, 89, 80, 90)
barplot(marks, names.arg = paste0("Student ", 1:5),
        main = "Marks of Students", ylab = "Marks",
        col = "lightblue")
barplot(marks, names.arg = paste0("Student ", 1:5),
        main = "Marks of Students (Horizontal)", xlab = "Marks",
        horiz = TRUE, col = "lightgreen")
```

OUTPUT:



RESULT:

The bar and horizontal bar plot was executed successfully.

EXPERIMENT-15

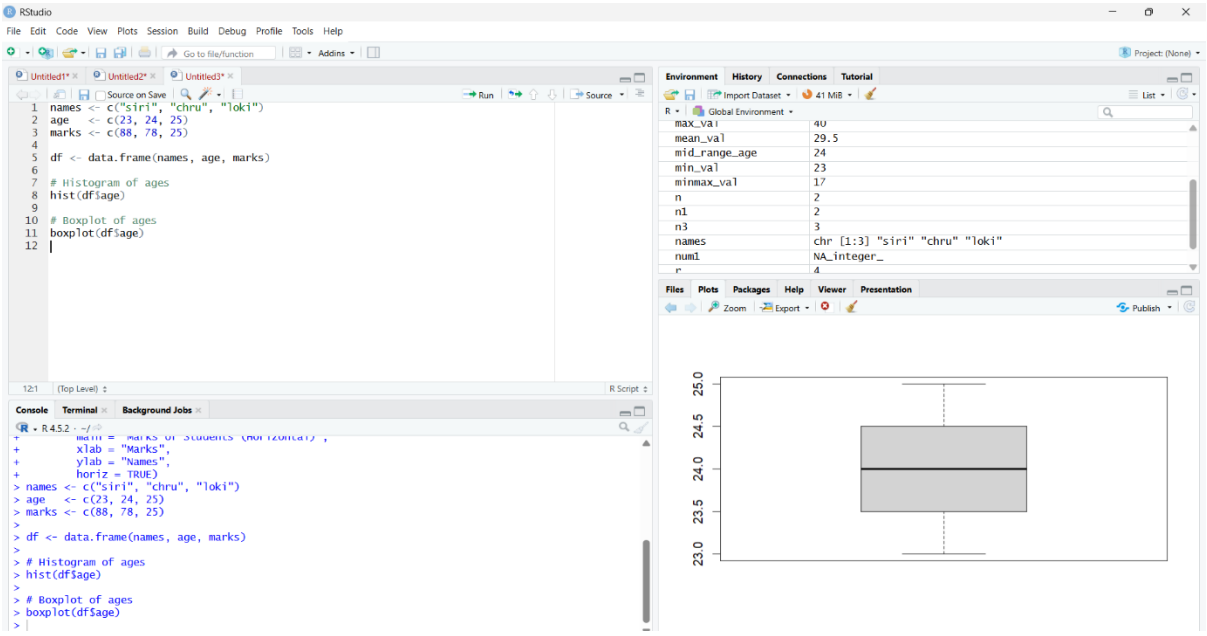
AIM:

To draw the box plot using R-tool.

PROGRAM:

```
names <- c("siri", "chiru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
boxplot(df$age, main = "Box Plot of Age", ylab = "Age")
```

OUTPUT:



RESULT:

The box plot was executed successfully.

EXPERIMENT-16

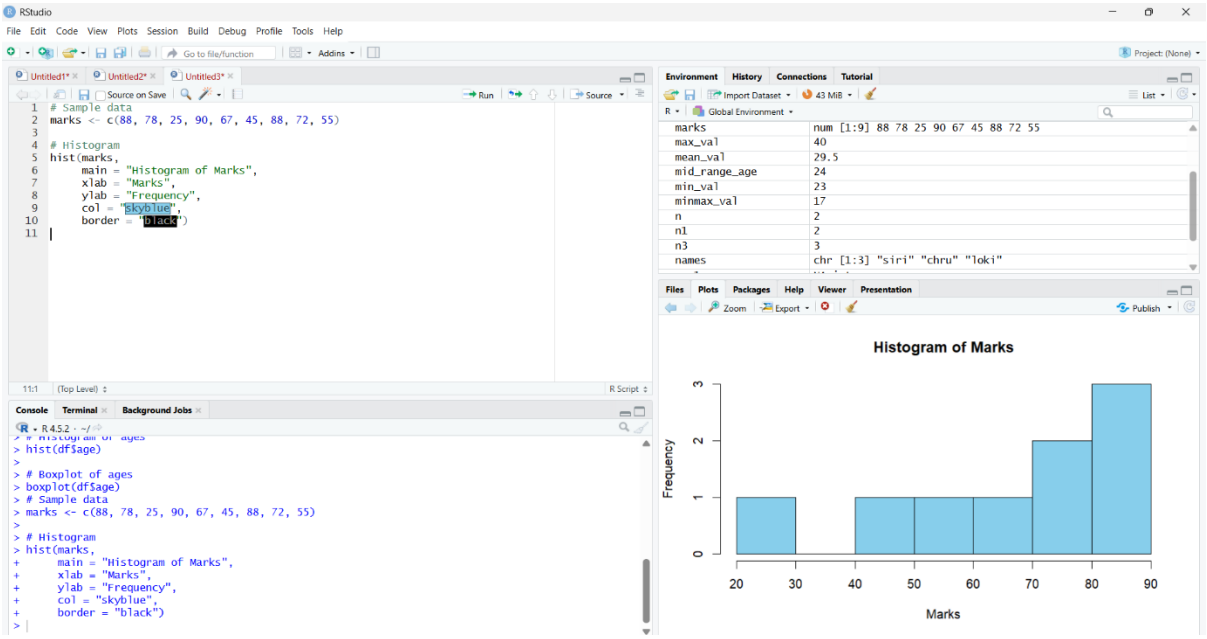
AIM:

To draw the histogram plot using R-tooll.

PROGRAM:

```
marks <- c(55, 67, 89, 80, 90)
hist(marks, main = "Histogram of Marks", xlab = "Marks",
     ylab = "Frequency", col = "lightblue", border = "black")
```

OUTPUT:



RESULT:

The histogram plot was executed successfully.

EXPERIMENT-17

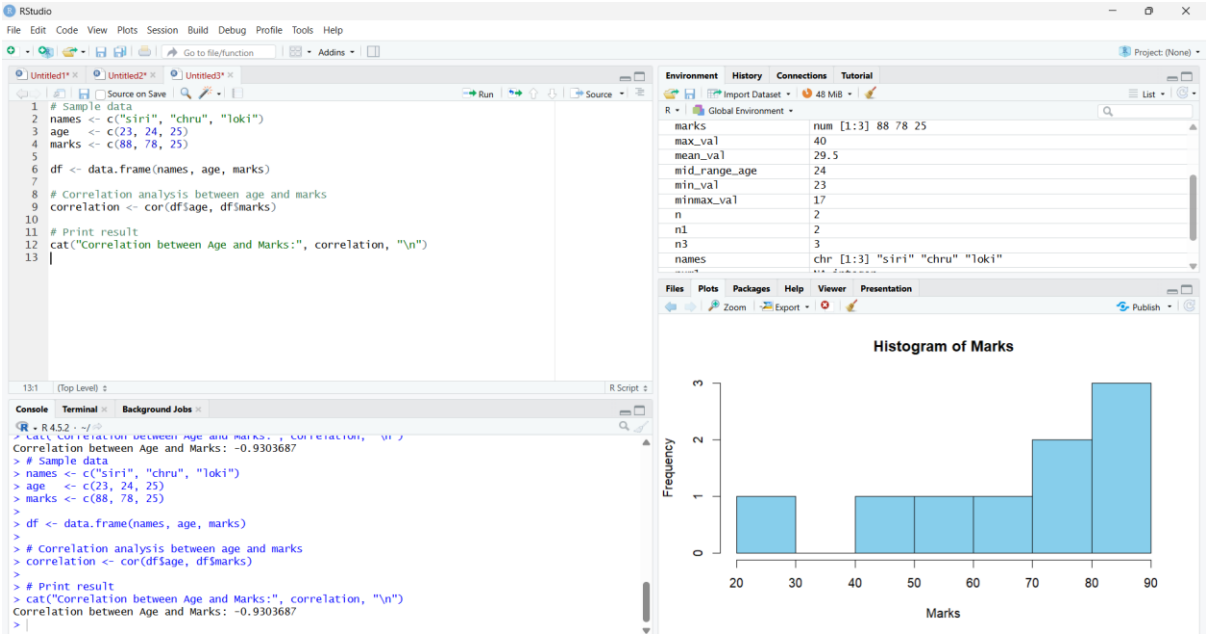
AIM:

To write the program for correlation analysis using R-tool.

PROGRAM:

```
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
correlation <- cor(age, marks, use = "complete.obs", method = "pearson")
print(correlation)
```

OUTPUT:



RESULT:

The correlation analysis was executed successfully.

EXPERIMENT-18

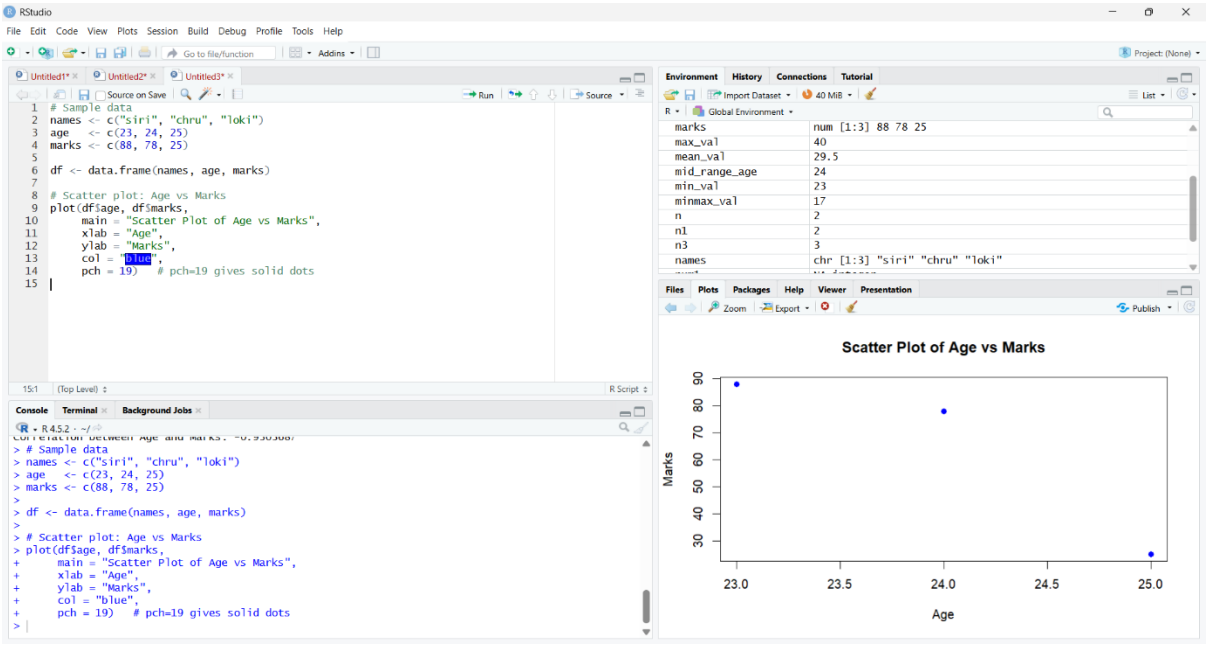
AIM:

To draw the scatter plot using R-tool

PROGRAM:

```
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
plot(age, marks, main = "Scatter Plot of Age vs Marks",
     xlab = "Age", ylab = "Marks", pch = 19, col = "blue")
```

OUTPUT:



RESULT:

The scatter plot was executed successfully.

EXPERIMENT-19

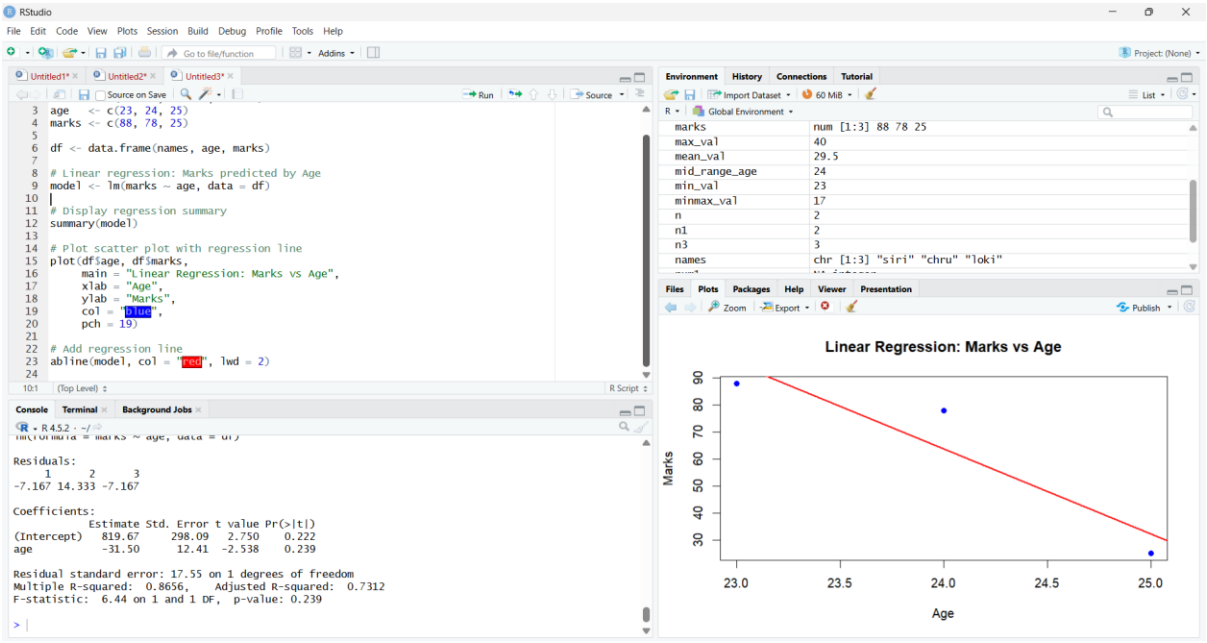
AIM:

To write the program for the linear regression using R-tool.

PROGRAM:

```
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
students <- data.frame(age, marks)
model <- lm(marks ~ age, data = students)
plot(students$age, students$marks, main = "Linear Regression: Marks vs Age",
     xlab = "Age", ylab = "Marks", pch = 19, col = "blue")
abline(model, col = "red", lwd = 2)
summary(model)
```

OUTPUT:



RESULT:

The linear regression program was executed successfully.

EXPERIMENT-20

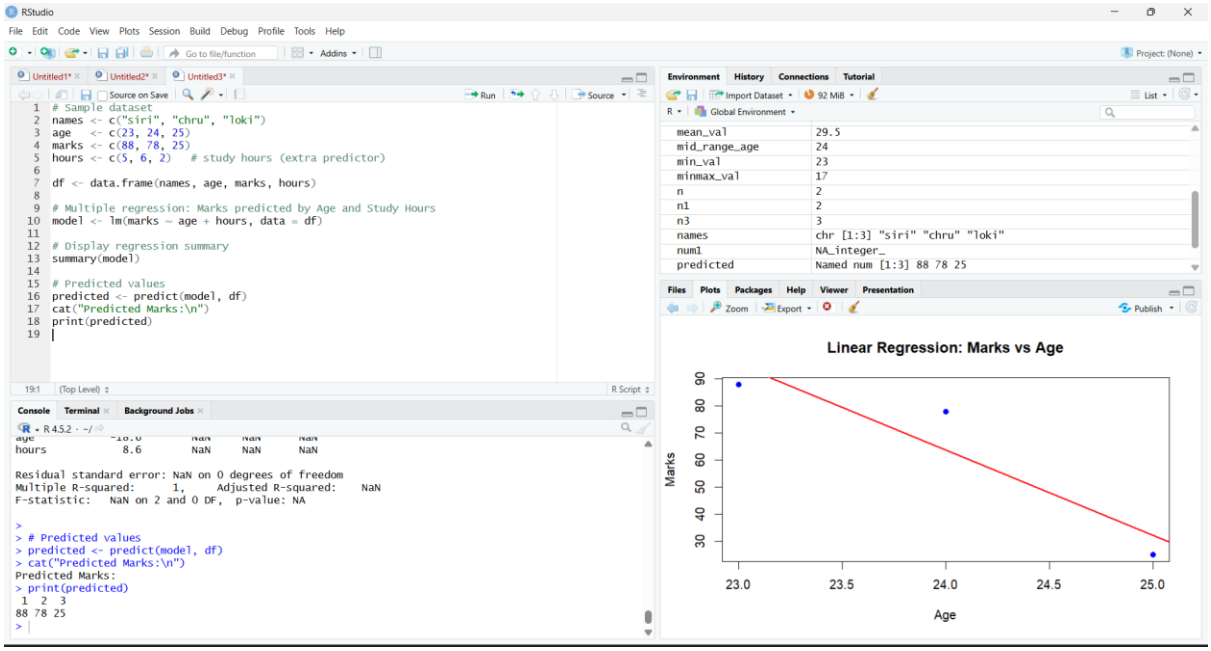
AIM:

To write the program for the multiple regression.

PROGRAM:

```
students <- data.frame(
  Age = c(20, 21, 22, 23, 24, 25, 26, 27),
  BloodPressure = c(110, 115, 118, 120, 125, 128, 130, 135),
  Glucose = c(90, 95, 100, 105, 110, 115, 120, 125)
)
model <- lm(Glucose ~ Age + BloodPressure, data = students)
summary(model)
# Optional: predicted glucose values
predict(model)
```

OUTPUT:



RESULT:

The Multiple Regression program was executed successfully.